



Smart nested origami-inspired detachable modular structures with tunable stiffness for multiphysics reconfiguration

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ABSTRACT

This study presents innovative architected metastructures that integrate nested origami with traditional lattice structures, addressing the limitations of fixed geometry and functionality in conventional lattice designs. Lattice structures exhibit high performance and lightweight design, enabling precise control of macroscopic physical properties through well-defined unit cell topologies. However, their post-manufacturing geometry and functionality are typically fixed, limiting adaptability in dynamic environments. The proposed novel metastructures achieve secondary expansion functions, termed “One-Double,” through modular design and reconfigurable mechanisms. The structures’ reconfigurability, detachable modularity, and tunable mechanical properties mimic the “metamorphosis” observed in biological systems, where organisms undergo significant morphological transformations. Building upon this, we propose three coupling-based reversible reconfiguration methods to enable dynamic geometry switching between different unit configurations. Experimental, theoretical, and simulation results demonstrate that these metastructures, due to their multifunctionality, show strong adaptability for satellite antennas and can guide fluid flow through reconfiguration, optimizing flow paths. This design approach paves the way for achieving complex multifunctionality across various domains, with significant potential for applications in dynamic adaptation, intelligent responsiveness, and functional integration.

1. Introduction

In recent years, lattice metamaterials [1–6] have emerged as a novel class of materials due to their unique microstructural designs. These structures exhibit distinctive properties, such as a negative Poisson’s ratio, tunable stiffness, and customizable thermal and acoustic behaviors, all stemming from the specific arrangement of their unit cells. They offer promising solutions for high-performance, lightweight structures [7,8] and highly integrated systems. Furthermore, by precisely controlling the topology, geometry, and size of the unit cells, these metamaterials enable the customization of macroscopic physical properties—capabilities that are typically not achievable by the constituent materials themselves. This fine control over the structure-property relationship is one of the key advantages of lattice metamaterials, allowing for the design of materials with tailored

performance characteristics. Due to these unique properties, three-dimensional lattice-based metamaterials have found widespread applications in aerospace [9,10], biomedical engineering [11], and the automotive industry [12]. For example, in aerospace applications, Zhang et al. [13] studied triply periodic minimal surface lattice structures for aerospace, finding the Diamond model excelled in heat transfer and insulation at low flow rates, while the Gyroid model performed better at high flow rates. In the field of biomedical engineering, the study by Li et al. [14] investigates the application of lattice structures in orthopedic implants. These three-dimensional printed lattice structures allow for the adjustment of mechanical properties to match the elastic modulus of human bone tissue, thus reducing the stress shielding effect. Additionally, the materials demonstrate good biocompatibility, showing promise for orthopedic repair. In the automotive industry, Zhang et al. [15] studied lattice structures with negative thermal expansion and

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negative Poisson's ratio for energy absorption. The configuration showed stable compression and high energy absorption, with negative thermal expansion helping reduce the initial thermal load. Both experimental and numerical results confirmed their effectiveness in impact protection systems.

The emergence of lattice structures undoubtedly marks a significant breakthrough in the field of structural engineering. However, engineers are increasingly recognizing the inherent limitations of highly integrated lattice structures manufactured through additive manufacturing or complex single-piece molding processes. For example, Li et al. [16] found that lattice structures made by additive manufacturing offer excellent mechanical performance and energy absorption, but their limitations include fixed geometry and functionality after production, making them difficult to adapt to changes in requirements, and complex maintenance due to the difficulty in repairing internal structures and potential unit cell damage. Zhang et al. [17] introduced a reconfigurable lattice structure with programmable deformation modes for advanced applications, but its reliance on fixed geometry after fabrication and the difficulty in repairing complex internal structures present challenges for long-term adaptability. Li et al. [18] introduced the hybrid lattice structure, which demonstrates effective vibration attenuation and energy absorption in the low and mid-frequency ranges. However, the structure's reliance on additive manufacturing for precise fabrication and its fixed geometric design pose limitations in terms of adaptability to functional adjustments and maintenance. In summary, the limitations of lattice structures are primarily reflected in the following aspects: First, once the lattice geometry and functionality are completed through additive manufacturing or complex single-piece molding processes, they become fixed and are difficult to adapt to changing demands. Second, the intricate internal design of the structure makes maintenance and repair challenging, particularly in the presence of localized damage or structural weaknesses. Lastly, although some designs possess reconfigurability or programmable deformation capabilities, they are still constrained by the fixed geometry, limiting their long-term adaptability and ability to adjust functionality.

Origami [19–22] has introduced the concepts of exceptional deformability [23] and reconfigurability [24], showcasing distinct advantages in the design of adaptive multifunctional materials. Origami designs exhibit anisotropic structural stiffness, enabling transformations between various shapes, such as contraction [25], extension [11], bending [26], torsion [27], and even complete changes in geometry. These properties make origami an ideal choice for applications requiring dynamic shape changes. Moreover, its dynamic deformation capabilities set it apart from traditional rigid structures, offering the flexibility necessary for real-time responses to external stimuli or constantly changing environmental conditions. For example, Zang et al. [28] investigated Kresling origami, demonstrating its ability to achieve programmable deformation through coupled axial displacement and twist, with potential applications in soft robotics. Zhao et al. [29] developed modular chiral origami metamaterials with multimodal deformation capabilities, including twist and contraction, offering flexibility for applications in robotics and energy absorption. Fu et al. [30] introduced an origami-based exoskeleton to enhance the stiffness and load-bearing capabilities of soft robotic arms, improving their performance in tasks like lifting heavy loads while maintaining flexibility and adaptability for pneumatic and cable-driven systems. In summary, origami designs, with their exceptional deformability and reconfigurability, demonstrate significant advantages in fields such as soft robotics, energy absorption, and multifunctional adaptive materials. Notably, they show considerable potential for widespread application, particularly in enabling dynamic shape changes, enhancing structural stiffness, and improving load-bearing capacity. Moreover, the multifunctionality of origami extends far beyond simple folding; it can also be used to design structures with variable stiffness, where different parts of the structure can dynamically adjust to meet specific functional requirements. A key example is the development of variable stiffness nested origami [31,32],

which enables multi-dimensional reconfiguration and adaptive performance. These origami structures exhibit significant potential in applications such as mechanical systems, automotive engineering, and energy dissipation devices [33–37]. The core idea of this paper is to further develop the concept of nested origami by exploring its multi-dimensional breakthroughs and reconfigurability potential, in conjunction with lattice metamaterials. The innovative and novel aspects of the metastructure are clearly illustrated in Fig. S1.

In this study, we combine the concepts of lattice metamaterials and nested origami to develop an origami-inspired demountable modular structure (ODMS), characterized by lightweight, high strength, reconfigurability, and adjustable stiffness. First, the accuracy of the simplified model was theoretically validated using finite element analysis (FEA) and the finite particle method (FPM). Next, numerical analysis methods (NAM) were employed to predict the node trajectories during the reconfiguration process. Experimental results and finite element analysis confirmed the changes in ODMS frequency, reflection, and transmission electromagnetic properties under external loading. Additionally, structural defects were introduced into the antenna model, and a machine learning-based mapping between defect characteristics and S-parameters was established [38–43]. To determine the optimal model for predicting S-parameter variations caused by defects, five algorithms were systematically compared to identify the most accurate prediction factor. Finally, wind tunnel experiments were conducted to visualize the flow field characteristics around various ODMS configurations, and experimental results verified the flow field predictions based on finite element analysis. FEA was also used to quantitatively assess the aerodynamic characteristics of each configuration, including drag, lift, and surface pressure distribution. The proposed system demonstrates reversible deformation, multifunctional adaptability, and multifunctional reconfigurability, enabling structural shape modulation in dynamic environments. These features highlight its vast potential for applications, particularly in satellite communication, adaptive flow control, and smart structures.

2. Results

For the reconfigurability of the origami-based detachable modular structure (ODMS), we identify three typical reconfiguration modes. The first mode, as shown in Fig. 1A, occurs when the external structure is under fixed constraint. By applying a linear displacement to the internal structure, the ODMS undergoes counterclockwise twisting (assuming a rotation angle of T1, displacement U1, and movie S1) during the reconfiguration process. This displacement-twist coupling characteristic arises from the geometric constraints and interactions between the internal structure and the external guide rails. The second mode, as shown in Fig. 1B, occurs when the internal structure is under fixed constraint. By applying a linear displacement to the external structure, the ODMS undergoes clockwise twisting (assuming a rotation angle of T2, displacement U2, and movie S2) during the reconfiguration process. The third mode, as shown in Fig. 1C, occurs when both the external and internal structures are released from constraints and undergo free reconfiguration. In this case, the two structures exhibit opposite rotations (assuming rotation angles of T3 and T4, displacements U3, U4 and, movie S3).

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As shown in Fig. 1E–D, fluid guidance is an effective method aimed at enhancing satellite antenna communication performance, particularly when low Earth orbit satellite antennas are affected by atmospheric drag and fluid variations. Fluid guidance technology can effectively optimize the flow field around the antenna, thereby reducing the impact of atmospheric disturbances on signal propagation. Based on this, it is necessary to clearly define and study the electromagnetic and fluid performance standards to ensure the effectiveness and reliability of the fluid guidance structure in practical applications (see Fig. S3).

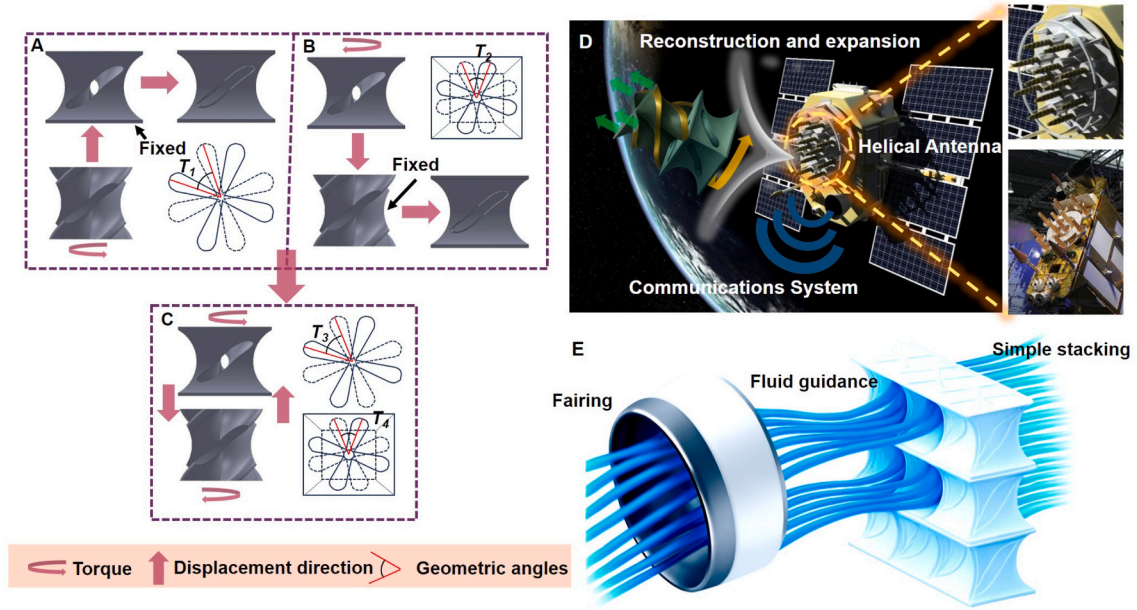


Fig. 1. Reconstruction of reversible methods and multi-field applications. (A) The gap between the inner and outer structures is calibrated, with the outer structure fixed in place. A displacement drive is then applied to the inner structure, enabling it to fully enter the outer structure and form the ODMS. At this point, the total height corresponds to the combined height of the outer and inner structure monomers. (B) The internal structure is fixed, and the external structure is driven into it through displacement to form the ODMS. After reconstruction, the total height remains equal to the sum of the heights of the internal and external structure monomers. (C) Once the internal and external structures of the ODMS are freed from constraints, they will automatically undergo reverse rotation based on their inherent kinematic trajectories during the relative displacement, ultimately forming a complete ODMS. (D) Application of helical antenna in satellite. (E) ODMS can be used as a flow guiding module to optimize the airflow into the engine or duct system.

The angle relationship can be expressed as

$$T_1 + T_2 = 2(T_3 + T_4) \quad (1)$$

The displacement relationship can be expressed as

$$U_1 + 2U_3 = U_2 + 2U_4 \quad (2)$$

According to the structural symmetry and linkage relationship

$$\begin{aligned} T_1 &= T_2 = 2T_3 = 2T_4 \\ U_1 &= U_2 = 2U_3 = 2U_4 \end{aligned} \quad (3)$$

2.1. Mechanically coupled reconfiguration and tunable stiffness of nested metastructures

This section is based on mechanically coupled metamaterials with intelligent reconfigurability. By reprogramming the configuration of both the inner and outer structural layers, the mechanical response can be effectively controlled, exhibiting highly coupled force–displacement and strain–torsion behaviors, thereby significantly enhancing the controllability and adaptability of the structure. We consider the soft elastic material (TPU) and its inherent structural properties, deriving a specific displacement–torsion coupling angle relationship from compression experimental analysis, from which typical parameters are selected (Fig. S1). In the following text, unless specified otherwise, $D = 2$ denotes a displacement of 2 mm and a rotation of 15° , $D = 6$ denotes a displacement of 6 mm and a rotation of 45° , and $D = 10$ denotes a displacement of 10 mm and a rotation of 90° .

The ODMS consists of two parts: the inner layer and the outer layer, which can be combined into a whole through reconstruction. To better characterize this process, we present the geometric parameters of each part and the overall reconstruction process (external fixation) in Fig. 2A. Furthermore, the load path, boundary conditions, and constitutive assumptions are detailed in Fig. S4; the modal stability analysis is presented in Fig. S5; mesh and parameter convergence are shown in Fig. S6; and the fixture constraints, loading protocol specifications, friction

calibration and specifications, as well as the friction sensitivity analysis, are demonstrated in Fig. S7.

As shown in Fig. 2B, we discretize the continuous physical process, the smooth trajectory curves further validate its excellent transition capability during the reconfiguration process. To accurately reveal the kinematic mechanism of ODMS during reconstruction and predict the node trajectory, we apply the kinematic formula along with the spline interpolation method [44], a multi-objective error function [45], and the Runge-Kutta fourth-order integration algorithm [46] (Table S1). As illustrated in Fig. 2C, the external and internal structures together form the Origami Detachable Modular Structure (ODMS). A comparison of the force–displacement curves of the ODMS and the external structure, based on both simulation and experimental data, reveals that the interaction between the internal and external structures of the ODMS results in more complex mechanical behavior. The coupling of the internal and external structures not only increases the system's stiffness but also amplifies the nonlinear force response. The rapid rise in the ODMS curve indicates that the system experiences higher stress concentration and localized plastic deformation during deformation, with the increase in force surpassing the nonlinear response of the single external structure. As shown in Fig. 2D, as the friction coefficient decreases from 0.7 to 0.4, the responses of friction force and contact stress become progressively smoother, with a reduction in both the amplitude and frequency of each peak. A higher friction coefficient causes abrupt changes in friction force and a larger nonlinear response, whereas a lower friction coefficient results in a smoother friction force distribution, indicating that the friction force response during displacement tends to become more linear and uniform.

Additionally, to more accurately characterize the complex deformation behavior of ODMS, we integrate experimental measurements with theoretical modeling. By measuring strain data during the actual compression process and simplifying the ODMS model based on the finite mass method, we enhance our understanding of ODMS deformation, while improving the efficiency and reliability of numerical analysis, providing guidance for the design optimization and functional

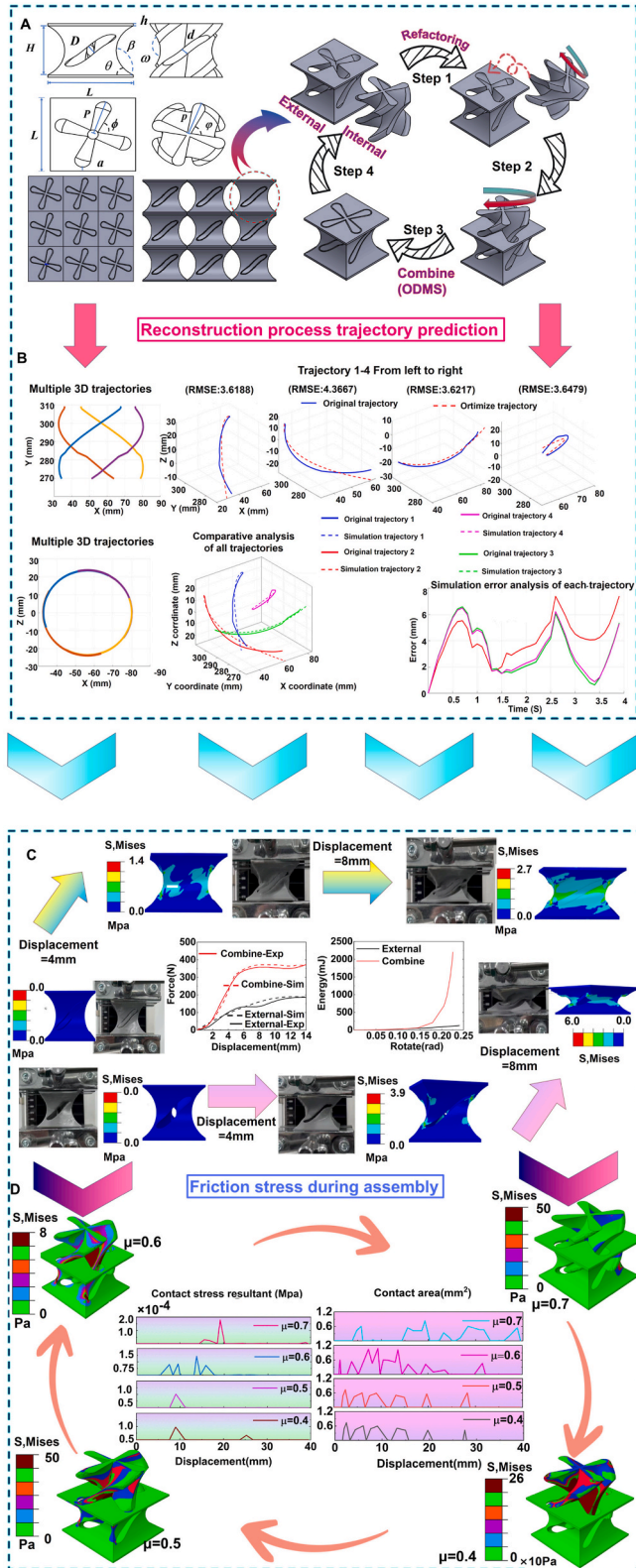


Fig. 2. Mechanical analysis. (A) Geometric parameters and reconstruction process. (B) Force-displacement and energy-angle relationships between ODMS and external structures under tension and compression tests. (C) Relationship between contact stress resultant force, contact area and displacement during ODMS reconstruction (external fixation). (D) ODMS node trajectory (external structure fixed, internal structure edge vertex) prediction and error analysis during reconstruction (Fig. S8).

regulation of ODMS.

The strain test results (Fig. 3A) reveal that when the external structure of the ODMS is subjected to compressive load, the critical region A (near the notch) experiences tensile stress. This localized stretching phenomenon is directly attributed to the torsional deformation that occurs during the downward compression displacement, which causes tensile effects in the material near the notch. Further analysis, based on the overall behavior of the ODMS, shows that as the downward compression displacement increases, the internal structure gradually fills the external structure and resists deformation. This internal filling effect effectively reduces the tensile strain in region A, thereby optimizing the overall stress distribution of the structure.

By analyzing the strain curve at point B, we observed that during the compression of the external structure of ODMS, the strain response at B

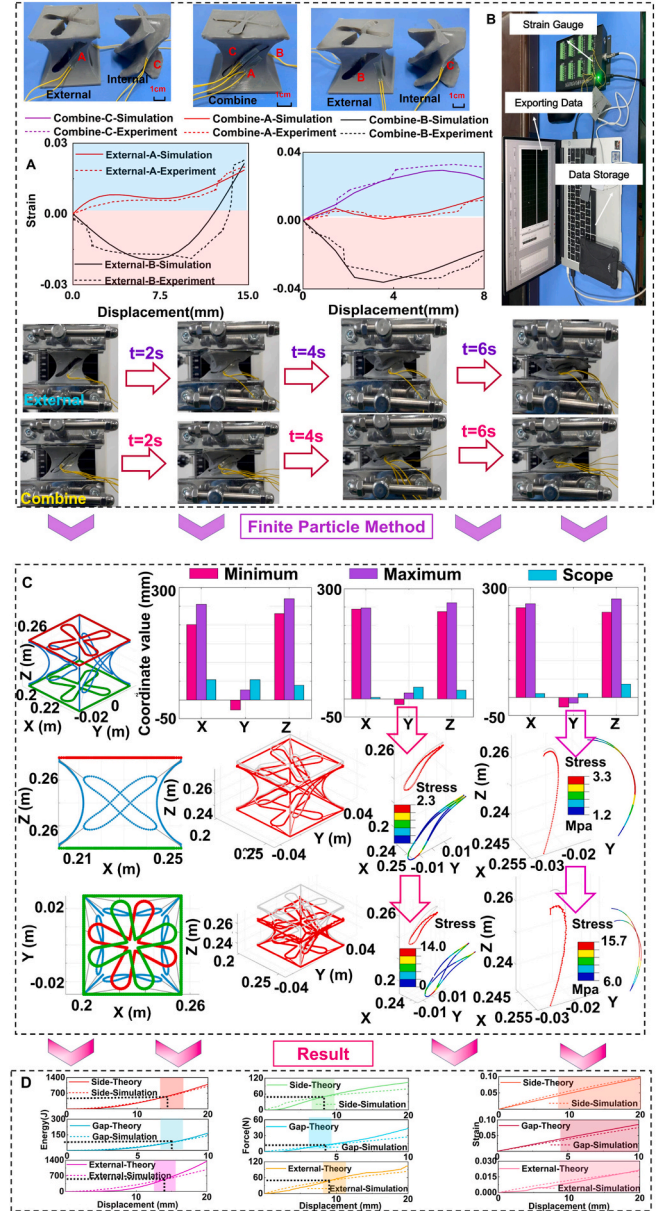


Fig. 3. ODMS deformation mechanism and simplified theory. (A) ODMS (Combine) strain gauge placement in compression test, comparison of experimental and simulation curves during deformation. (B) Instruments used in the experiment. (C) Using the finite mass method, the model simplifies the node connection, node coordinate distribution, deformation process, and stress analysis under FEA. (D) Comparison of the accuracy of FEA and theoretical curves (energy, force, strain-displacement).

exhibited a complex multi-stage variation. In the initial stage, strain increased gradually due to the displacement-torsion coupling effect, leading to a steady intensification of strain. After reaching a certain level of torsion (when the upper and lower walls of the notch were not in contact), the strain began to decrease. Finally, with continued displacement loading, the upper and lower walls of the notch came into contact, subjecting point B to vertical extrusion and inducing compressive strain. Comparing the strain curve at point B of the entire ODMS, we

found that the introduction of the internal structure mitigated the drastic strain variations during compression and effectively dispersed local stress concentrations. This also confirmed the accuracy of the strain curve at point A.

To better understand the complex deformation behavior of ODMS, we apply the theory of the Finite Particle Method [47,48], where each node of the ODMS corresponds to a particle, and its coordinate distribution (particle distribution) is shown in Fig. 3C.

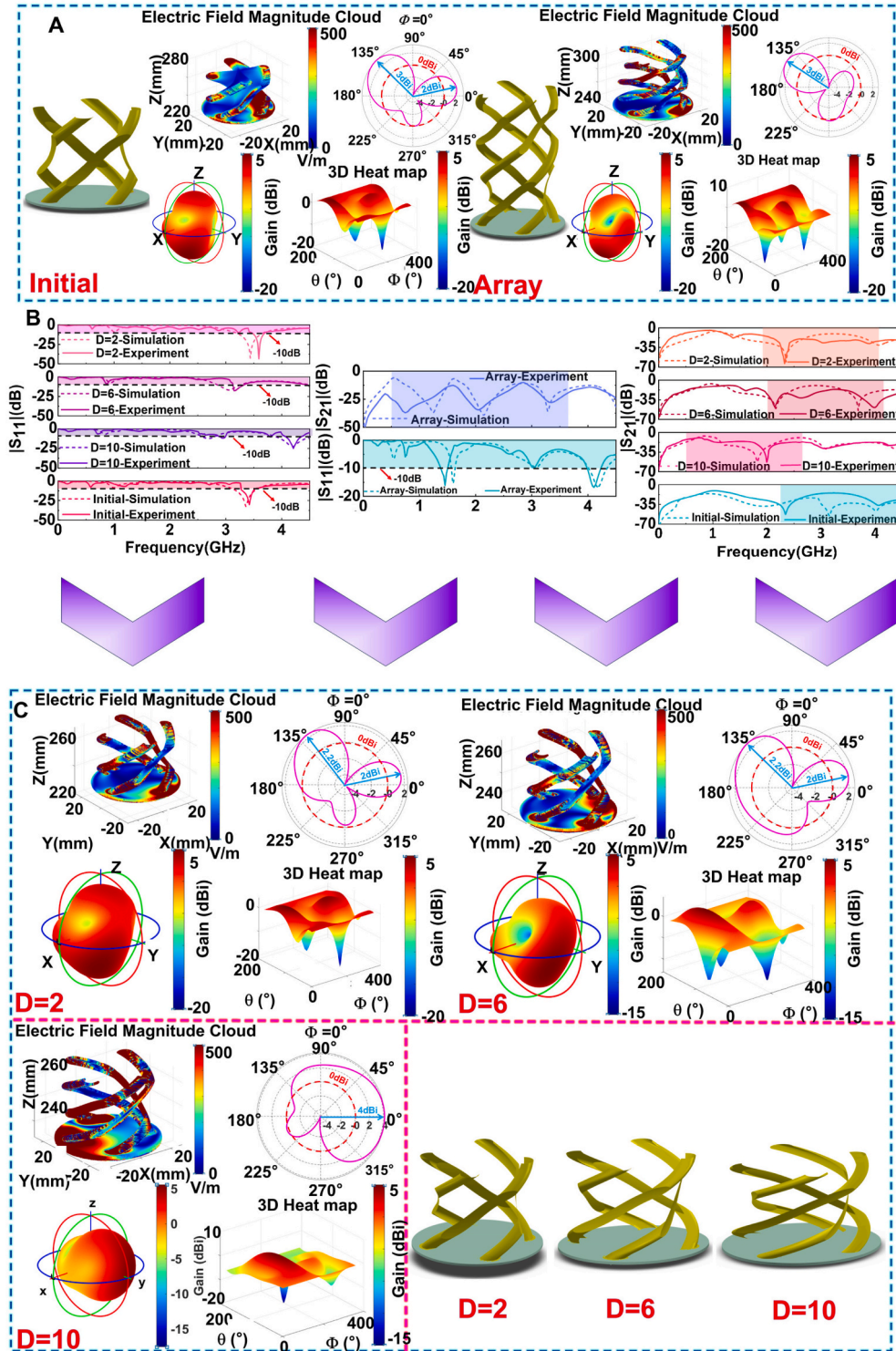


Fig. 4. Electromagnetic properties. (A) Electromagnetic performance of ODMS antenna in initial configuration and longitudinal array configuration. (B) Comparison of reflection and transmission curves in various states. (C) Electromagnetic performance of ODMS antenna under different deformation configurations.

At the lowest point of the bottom (Z-axis), we choose to fix it at point $\vec{r}_j = \vec{r}_{j,0}$, $\vec{v}_j = \vec{0}$, $\vec{a}_j = \vec{0}$, allowing only movement along the Z-axis, denoted as $x_j = x_{j,0}$, $y_j = y_{j,0}$, $v_{xj} = 0$, $v_{yj} = 0$, $a_{xj} = 0$, $a_{yj} = 0$.

In addition, we choose to apply the force to the top mass point (the highest point on the Z axis), and the force distribution follows $F_{\text{per,ode}} = \frac{F_{\text{total}} \times \alpha(t)}{N_{\text{top}}}$. At the same time, in order to avoid impact, a progressive

loading strategy $\alpha(t) = \min\left(1.0, \frac{t}{0.1 \times t_{\text{total}}}\right)$ is adopted to gradually apply the load in the first 10 % of the time.

Through result analysis, we found that the overall trend of the curves remains consistent, following the principle of local-to-global behavior, where the mechanical behavior of the micro-units accumulates to form the macroscopic response of the structure. Additionally, the mechanical curve of the supporting edge closely resembles that of the external structure, indicating that under loading conditions, the supporting edge is one of the primary load-bearing components, experiencing loads similar to those of the overall external structure.

2.2. Morphology-driven electromagnetic reconfiguration for adaptive radiation control

This section demonstrates the dynamic tunability of the electromagnetic properties of the metamaterial (ODMS) under morphological transformations. It achieves electromagnetic beam shaping and frequency tuning from the unit-cell state to the array state, highlighting its potential for multifunctional applications such as reconfigurable antennas. Above, we have explored the complex nonlinear mechanical behavior of ODMS under external loads, including its unique displacement-torsion coupling deformation mechanism and the role of the internal structure in strain. Next, based on the morphological changes caused by these mechanical deformations, we will further analyze how ODMS regulates its response to electromagnetic waves through morphological reconstruction. Using 3D printing technology, we printed ODMS in different configurations and integrated metal patches onto the non-Euclidean outer surface of the internal structure of ODMS, thereby constructing RF antennas in different forms [49–51] (Fig. S9). In this paper, the physical meaning and calculation formula of the S-parameters are detailed in Fig. S10.

By analyzing the radiation patterns and directional diagrams (Fig. 4A) of the ODMS antenna in both the initial and array configurations, we observed significant differences in the radiation patterns between the two states. The main lobe gain of the initial and array patterns is primarily concentrated in the direction of $\theta = 135^\circ$ when $\Phi = 0^\circ$. The main lobe angular difference is approximately 10° , resulting in a gain of 3 dBi. Based on $G = 10 \times \log_{10}\left(\frac{P_{\text{real}}}{P_{\text{ref}}}\right)$, the radiation intensity is approximately double that of an ideal antenna. In the initial state, the antenna's radiation intensity at $\theta = 20^\circ$ is 1.6 times the power of an ideal antenna, while in the array state, it is double the power level. Furthermore, the main lobe gain of the antenna in the array state is more concentrated than in the initial state, with the overall beam angle θ shifting 10° higher compared to the initial state. At $\theta = 30^\circ$, the sidelobe radiation intensity is reduced by 2.5 dBi. The 2D directional diagram shows a more divergent beam, while the 3D gain heatmap visually illustrates the energy distribution across a broader solid angle. However, when the ODMS units are mechanically reconfigured into an array configuration, the radiation pattern shifts to a narrower main lobe with highly concentrated energy and a higher peak gain. At this point, the 2D directional diagram clearly shows a significantly converging beam, and the 3D gain heatmap also demonstrates the focusing of energy toward a specific direction. This dynamic switching of the beam shape from wide coverage to high directionality directly demonstrates the ability of ODMS to effectively control electromagnetic radiation direction and gain through simple array configurations. In addition, circular polarization validation and reproducibility testing are provided in

Fig. S11–S12.

In the experiment, we also considered integrating conductors into the inner surface of the external structure and connecting them to the actual antenna (as shown in Fig. S13, where the conductors are integrated into the outer surface). However, it should be noted that the electromagnetic performance tests did not show significant effects. But it demonstrates the potential for evolving from a conventional static antenna structure to an intelligent configuration with electromagnetic dynamic reconfigurability. 2D radiation heat map of all types of antennas (Fig. S14)

As shown in Fig. 4C, we observe discrete and significant reconfigurations in the antenna's beam shape, width, and main lobe direction as the deformation degree changes from $D = 2$ to $D = 6$ and then to $D = 10$. The antenna initially shows an increase in side lobe gain, followed by a noticeable adjustment in the radiation main direction. In fact, we consider this driven by the torsion angle accompanying the height reduction of the ODMS antenna, which causes stretching and growth of the internal conductors. In the initial stage, slight torsion and stretching disturb the antenna's original current distribution, activating or enhancing some weak current paths, which results in an increase in side lobe gain. As displacement increases further, the torsion angle and stretching become more pronounced, leading to a fundamental reconstruction of the internal current distribution pattern of the antenna. This fundamental reconstruction enables the antenna's main radiation lobe to clearly shift from one direction to another, achieving a noticeable adjustment in the main radiation direction, manifested as beam shaping or steering through mechanical deformation.

Additionally, the analysis of S-parameters is crucial in antenna design and performance evaluation. As shown in Fig. 4B, S-parameters for reflection and transmission performance were obtained for different states. In terms of reflection performance, comparing the initial state with $D = 2$, $D = 6$, and $D = 10$, we observed that the reflection coefficient gradually increased, with multiple small peaks, and the main resonant frequency exhibited an upward drift. Additionally, comparing the initial state with the array state, we found that the S11 curve in the array state displayed multiple resonance peaks, with the reflection coefficient consistently below -10 dB within the target frequency range. The ODMS antenna, through simple mechanical deformation, effectively achieves an upward tuning of its operating frequency, accompanied by an increase in broadband impedance matching complexity.

2.3. Machine learning-enabled robustness evaluation under multi-parameter structural defects

By incorporating machine learning techniques, an efficient mapping between structural defects in the metamaterial and its electromagnetic performance was established, demonstrating a key advantage: the ability to maintain performance predictability and robustness even in the presence of microstructural damage. Over extended operational periods, the antenna surface inevitably experiences corrosion as time progresses [52–54]. Due to the complexity of corrosion phenomena, its description often involves multiple parameters (such as corrosion quantity, diameter, and depth). In such complex multi-parameter cases, traditional analysis methods struggle to comprehensively assess the combined effects of corrosion on antenna performance. Therefore, by integrating and comparing various machine learning algorithms, we analyze multi-parameter corrosion data to predict and quantify its impact on antenna electromagnetic performance, while developing a model that effectively characterizes the complex relationship between corrosion and ODMS antenna performance. Furthermore, the results of this study exhibit significant generalizability. To further validate this, cross-validation protocols (see Fig. S15), distribution shift tests (see Fig. S16), and feature importance and partial dependence analyses (see Fig. S17) were conducted. These validation efforts ensure the broad applicability of the results and their stability under different conditions.

We divide the characterization of corrosion defects into five pa-

rameters (Fig. 5A), namely: number of defects, minimum radius of defects, maximum radius of defects, minimum depth of defects, and maximum depth of defects. It should be noted that the distribution of defects is based on normal distribution $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, which is the method we adopted after weighing the practical, simplified analysis and easy control. In addition, we divide the sample data into training set, test set, and validation set, of which 60 % is used as training

set, 20 % as test set, and 20 % as validation set (Fig. 5C).

Through training, we found that the Random Forest model performs well in predicting the relationship between defect features and ODMS antenna electromagnetic performance under the same conditions. As shown in Fig. 5D, in the comparison plot of the best model (RF) predictions, most of the blue prediction points are tightly distributed near the red ideal prediction line and the green fitted trend line, with $R^2 = 0.8181$ and $RMSE = 3.65$, both within acceptable ranges. It should be

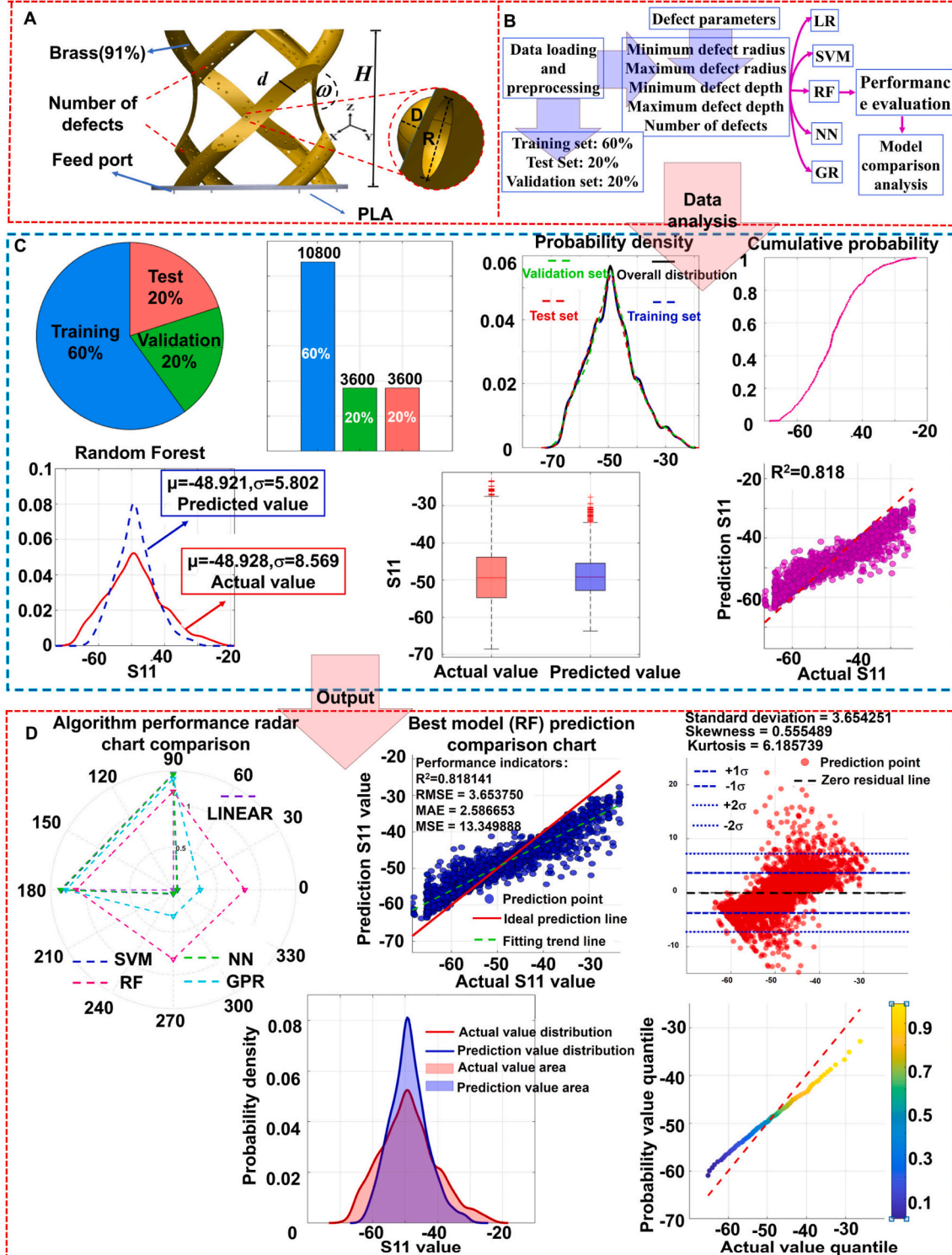


Fig. 5. Feature analysis. (A) Schematic diagram of antenna surface defects. (B) Main flow chart. (C) Data partitioning and multiple algorithm analysis (Fig. S18). (D) Comparison of algorithm prediction effects and final prediction performance.

noted that in the residual analysis plot, we observed greater dispersion of the residual points in the lower and higher parameter regions, indicating some limitations in the model's prediction of extreme values. However, the overall performance of the model is sufficient to support its application in analyzing the impact of surface defects on antenna electromagnetic performance. Additionally, we plotted the distribution of predicted values versus actual values and the comparison of predicted and actual value percentiles. Through analysis, we found that the overall statistical distribution closely matches the actual S11 value distribution, effectively capturing the overall data characteristics while accurately reflecting the range and probability distribution of the S11 values.

It is important to emphasize that one of the core goals in designing the ODMS antenna was to achieve high controllability and robustness in both structure and function. Interestingly, when predicting the complex relationship between defects in the ODMS antenna and its electromagnetic characteristics, we found that no overly complex modeling algorithms were necessary. The Random Forest model alone was sufficient for effective prediction. This not only simplified the modeling process and reduced computational costs but, more importantly, it validated that the key electromagnetic characteristics of the ODMS antenna could still be captured and predicted efficiently and accurately, even in the presence of defects. This result perfectly aligns with our initial goal in creating this reconfigurable antenna, which was to ensure its adaptability and performance predictability in practical applications.

2.4. Flow-guiding mechanics and vortex modulation via structural morphing

This section analyzes the coupling effects between the material structure and the flow field under multiple morphological configurations. It reveals that the structure can precisely guide airflow, generate shear layers and vortex streets through reconfigurable shapes, exhibiting intelligent capabilities in flow control and environmental responsiveness. In addition to the analysis of ODMS mechanical behavior and electromagnetic characteristics, we recognize that the fluid dynamic environment encountered when ODMS operates in air or fluid media is crucial to its performance and reliability. Based on this, we conducted a flow field analysis under different states, primarily using FEA and wind tunnel visualization for quantitative analysis.

It should be emphasized that this article simplifies the expressions in the legend to a certain extent. C represents Combine, referring to the overall structure (ODMS), and I represent internal, referring to the internal structure. E represents External, referring to the external structure, H represents Horizontal, referring to the horizontal flow field, and V represents Vertical, referring to the vertical flow field. D = 2 represents a vertical displacement of 2 mm with a 15° rotation, D = 6 represents a vertical displacement of 6 mm with a 45° rotation, and D = 10 represents a vertical displacement of 10 mm with a 90° rotation.

The wind speed was set to 50 m/s. Analysis revealed that in Fig. 6A, the streamlines around the internal structure of ODMS in the vertical flow field are distributed according to its geometric shape. When the fluid contacts the front-facing windward surface, the conversion of kinetic energy to pressure energy leads to a significant increase in pressure in this region, forming a stagnation point effect. Subsequently, the fluid is guided along ODMS's unique twisted and concave structure, where its velocity significantly increases as it passes through narrow internal or edge channels. At this point, $P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$ leads to $v_2 > v_1$, creating a local negative pressure zone. A clearer comparison with Fig. 6B shows that when the fluid passes over sharply changing curved surfaces of the structure, streamline separation occurs, leading to the formation of complex wake vortices behind the structure and in the recessed regions.

Additionally, we found that during the reconfiguration process, the highest-pressure region is not located, as traditionally understood, only at the front-facing windward surface of the structure, but is instead

concentrated on the external surface of the ODMS internal structure (non-vertical windward face). As the fluid flows along its complex internal twisted structure, local stagnation and significant deceleration occur in this region due to fluid compression and sharp turns, forming a pressure peak zone. Through extensive testing, we also observed an interesting phenomenon: the fluid is cleverly guided along the twisted shape of the internal structure and impacts the upper surface of the ODMS external structure. This process resembles the “dam effect,” where the redirected fluid directly impacts and vigorously merges with the high-speed edge fluid of the external structure, forming a strong shear layer. At the boundary where the fluid converges, a velocity difference ΔU exists, leading to $\omega_i \propto \frac{\Delta U}{L}$. The disturbance growth rate ω_i is proportional to the velocity difference. The larger the velocity difference, the faster the disturbance growth, promoting the formation of a distinct vortex structure at the gap. Notably, as the reconfiguration process progresses, the local impact and fluid convergence caused by the “dam effect” gradually diminish.

As shown in Fig. 6D, under vertical wind conditions, structural deformation leads to the formation of distinct vortex structures in the gap region, primarily caused by streamline separation when the fluid flows over sharply changing geometric features. Notably, as the deformation increases (from D = 2 to D = 10), the vortices gradually shift towards the rear of the structure, indicating that deformation not only affects vortex generation but also alters their spatial position and wake characteristics. In contrast, as shown in Fig. 6E, under horizontal wind conditions, the structure generates almost no noticeable vortices, reflecting the more streamlined aerodynamic profile of the ODMS in this flow direction, allowing the fluid to pass more smoothly and avoid large-scale separation.

3. Discussion and conclusion

We introduce an innovative metastructure concept featuring adaptive morphological transformation and intelligent control pathways. Integrating nested origami principles with a modular lattice, the system leverages geometric reconfiguration, mechanical coupling, and multi-scale regulation to jointly modulate structural, mechanical, electromagnetic, and fluidic behaviors—achieving intelligence and multifunctionality beyond conventional structures. A distinctive capability is the dynamic coupling of morphology, functionality, and environment via configurational reassembly, allowing stress-driven performance adaptation, analogous to organismal inside-out transformation in nature. Owing to its highly integrated and reconfigurable architecture, the platform shows strong potential for complex scenarios, notably in reconfigurable antennas, fluid-directing structures, and self-healing smart structures.

Our investigation reveals a displacement–torsion coupled deformation mechanism in the origami-inspired deformable metastructure (ODMS), where interactions between internal and external structures enhance stiffness and amplify nonlinear responses. We quantify the influence of friction on stability and mechanical behavior and develop a simplified model that effectively predicts ODMS deformation. In electromagnetic analyses, mechanical reconfiguration enables upward frequency tuning and dynamic beam-shape control, steering beams from wide coverage to high directivity. For antenna corrosion defects, a lightweight machine-learning model accurately captures the defect–performance relationship without complex modeling algorithms, greatly reducing computational costs. Regarding fluid dynamics, comparative tests under vertical and horizontal flows show that vertical flow induces distinct vortex structures and alters wake characteristics, whereas horizontal flow yields a more streamlined aerodynamic profile. These results underscore the ODMS's ability to modulate fluid behavior via morphological change and demonstrate adaptability and robustness across complex scenarios.

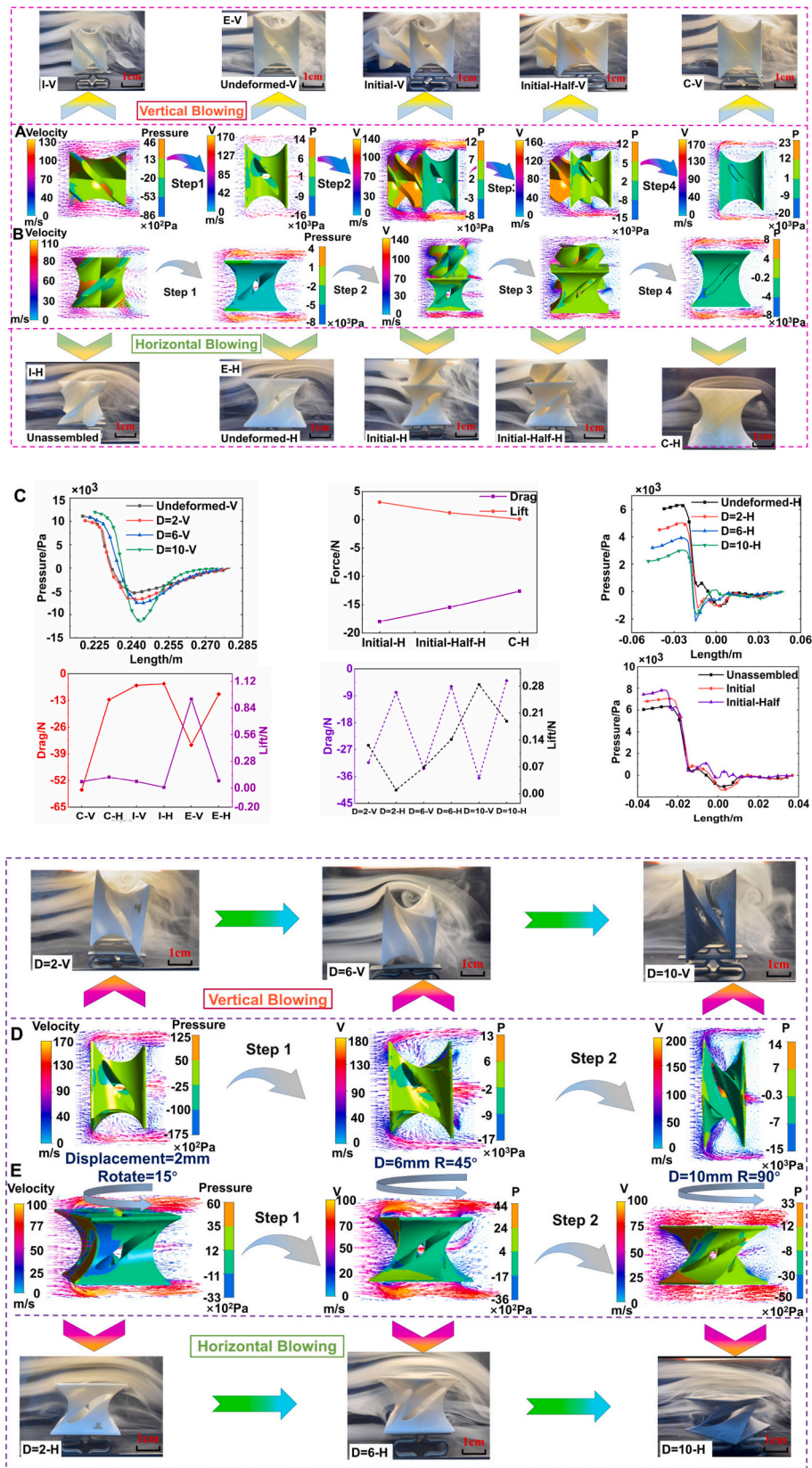


Fig. 6. Fluid analysis. (A) Vertical flow field during reconstruction (Fig. S19). (B) Horizontal flow field during reconstruction. (C) Lift, drag, and pressure curves under different conditions. (D) Vertical flow field of the ODMS external structure in a deformed state. (E) Horizontal flow field of the ODMS external structure in a deformed state.

CRediT authorship contribution statement

Ji Zhang: Writing – original draft, Funding acquisition, Resources, Supervision, Validation, Investigation, Formal analysis, Conceptualization, Data curation, Methodology, Software, Writing – review & editing. **Jintong Chen:** Writing – original draft, Methodology, Data curation, Formal analysis, Software, Writing – review & editing. **Shuai Liu:** Software, Data curation, Formal analysis, Investigation, Methodology. **Changguo Wang:** Writing – original draft, Funding acquisition, Conceptualization, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compositesb.2025.113164>.

Data availability

Data will be made available on request.

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